

Why MLA: one shared latent instead of 128 K/V pairs

KV cache per token

Both bars count the values cached for ONE key position — a single index of S_k , shared by all 128 heads and by every one of the S_q queries.

standard MHA — 128 heads $\times$ (192 K + 128 V) = 40,960 values per S_k position



→ MLA — $c_{kv} (512) + pe (64) = 576$ values per S_k position

The absorbed path: move the per-head matrix onto the query side

$$q_{nope}[h] : S_q \times 128 \quad c_{kv} : S_k \times 512 \quad W_{uk}[h] : 128 \times 512 \quad \text{so } K_{nope}[h] = c_{kv} W_{uk}[h]^T : S_k \times 128$$

$$q_{nope}[h] \cdot K_{nope}[h] = q_{nope}[h] \cdot (c_{kv} W_{uk}[h]^T) = (q_{nope}[h] W_{uk}[h]) \cdot c_{kv}$$

|
reconstruct K for all 128 heads
 $S_k \times 128 \times 128$ values
grows with the cache

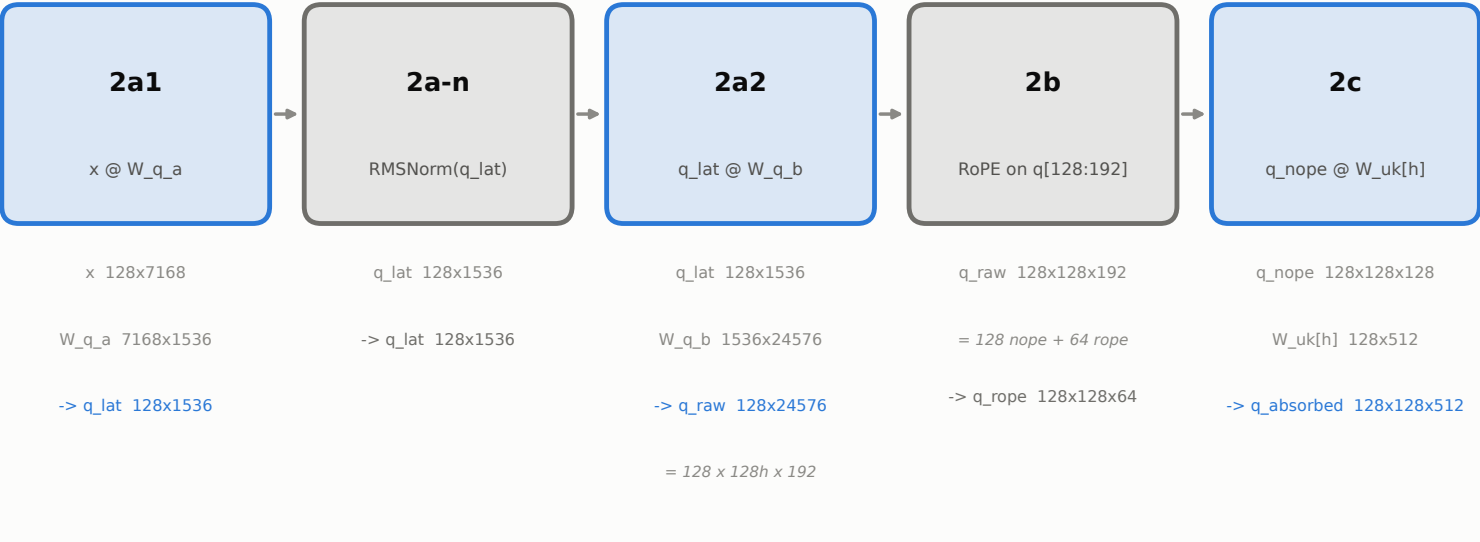
|
absorb into the query, ONCE
 $S_q \times 128 \times 512$ values
 $S_q = 1$ in decode

c_{kv} and pe_cache carry no head index: all 128 heads of a query read the SAME cache rows, so kernel 3a gives one block all 128 heads at once.

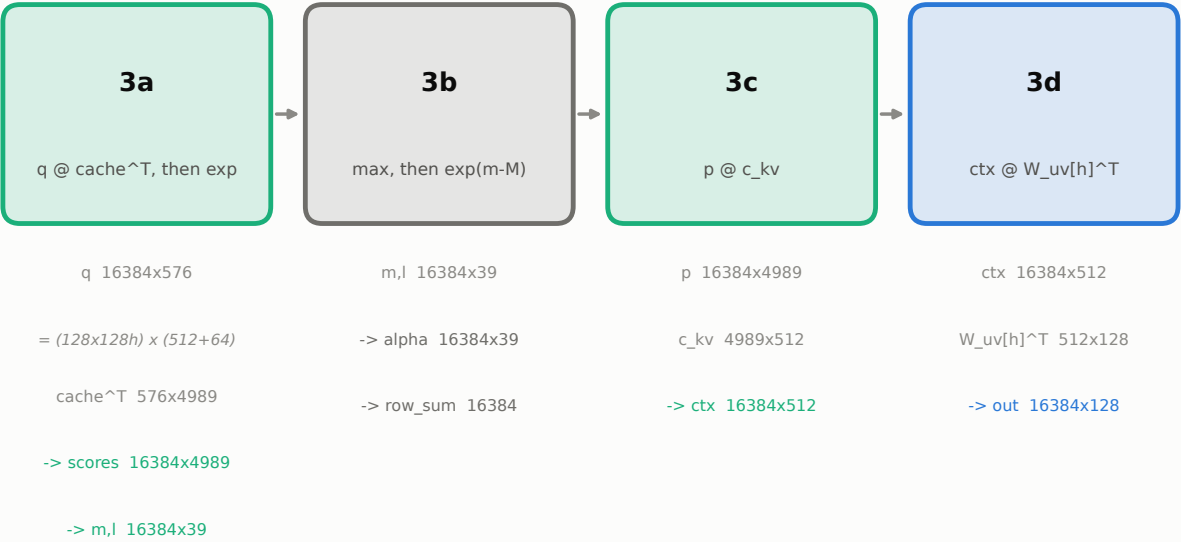
The 9 steps

Kernel 2 builds the 576-wide query; kernel 3 does the attention. Shapes for batch-128 decode: bq = B.Sq = 128 rows, flat = B.Sq.H = 16,384 rows, Sk = 4,989. Every operand is listed by name; outputs are marked ->.

KERNEL 2 — Q path



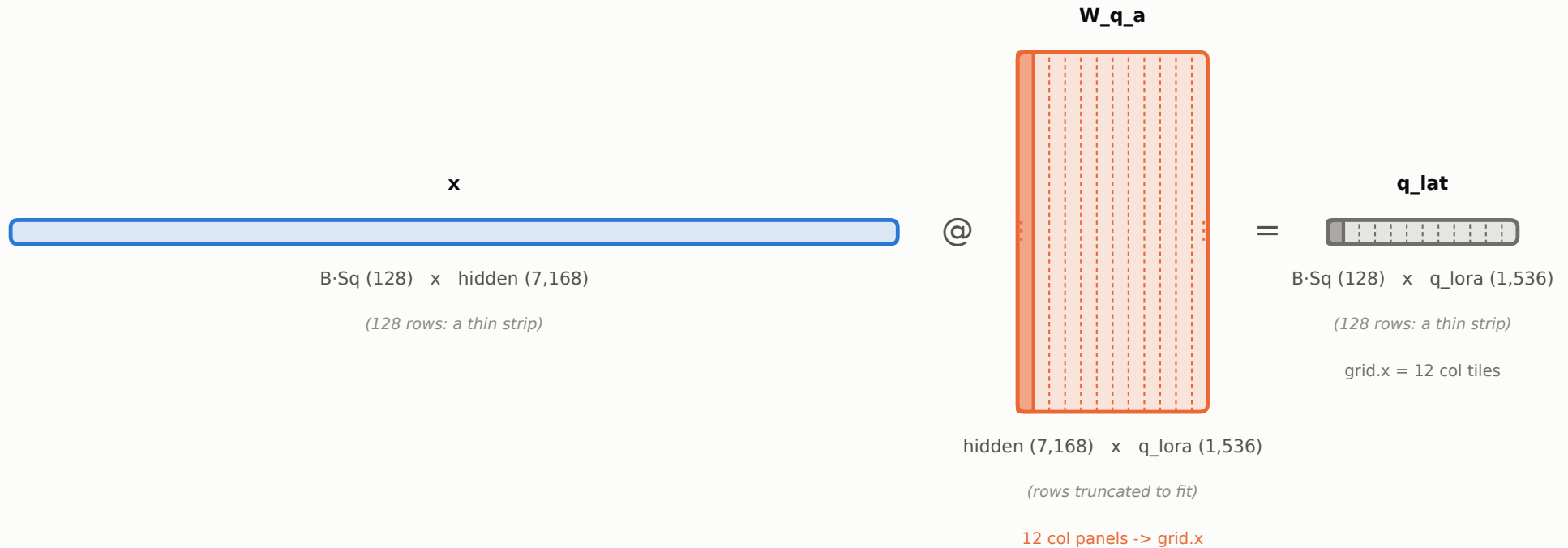
KERNEL 3 — attention



Step 2a1 — project the hidden state into the q_lora latent

$$q_lat = x @ W_q_a$$

first half of the factorised Q projection. 7,168 = hidden. 1,536 = q_lora_rank.



grid.x -> 12 output-column tiles of 128 | grid.y -> 1 row tile (bq = 128) | grid.z -> split-K: the 7,168 reduction is cut into slices
Only 12 column tiles means 12 blocks on an 84-SM GPU — split-K is what makes this step use the machine at all.

Step 2a-norm — RMSNorm on the latent

$q_lat := q_lat * \text{rsqrt}(\text{mean}(q_lat^2) + \text{eps}) * \text{gain}$

row-wise; keeps the two matmuls from collapsing into one rank-1536 matrix

q_lat (in and out)

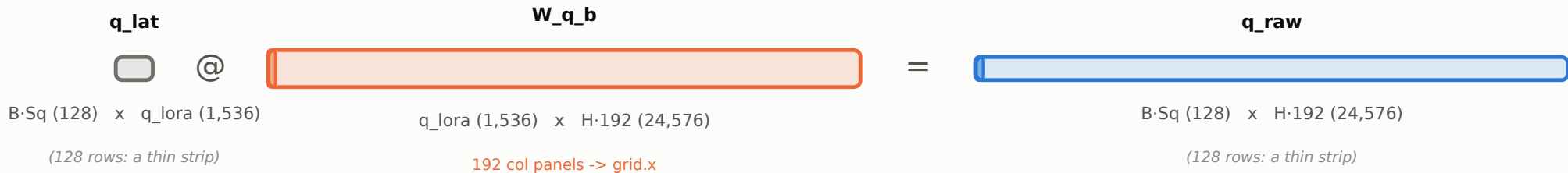


B·Sq (128) x q_lora (1,536)

one CUDA block per row -> 128 blocks
256 threads cover 1,536 elements (6 each)
each warp of 32 shuffles down to one number
warp 0 then sums those 8 numbers into the total

Step 2a2 — expand the latent into the per-head query

$$q_{\text{raw}} = q_{\text{lat}} @ W_{q_b} \quad (24,576 = 128 \text{ heads} \times 192)$$



grid.x -> 192 output-column tiles of 128 | grid.y -> 1 row tile | grid.z -> split-K over the 1,536 reduction
192 blocks against 84 SMs x 2 resident = 168 slots is 1.14 waves: one full wave, then a tail wave leaving 144 slots idle.

grid.x = 192 col tiles

Step 2b — RoPE on the rope half of each head

$q_rope[b,s,h,:] = \text{rotate}(q_raw[b,s,h, 128:192], \text{position}[s])$

elementwise; rotates adjacent pairs by an angle set by the query's ABSOLUTE position

q_raw, ONE head: 192 values

nope [0:128] -> step 2c

rope [128:192]
rotated here

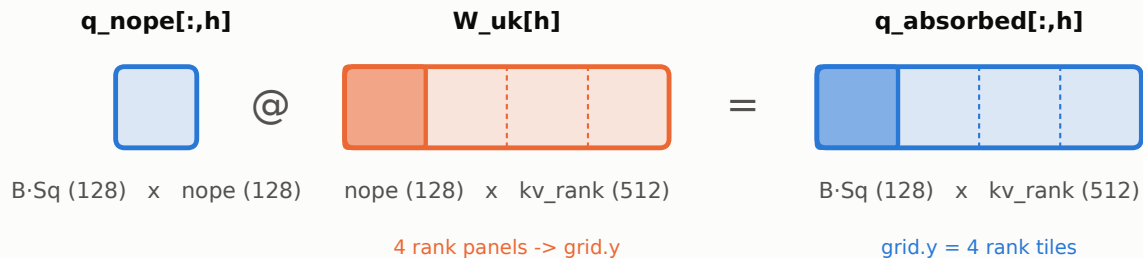
over all rows and heads: [128 x 128 x 192] -> q_rope [128, 128, 64]

One thread per (row, head, adjacent pair), grid-stride capped at 32 x SM so the launch never scales with the cache depth.

Step 2c — absorb W_{uk} into the query (once per head)

$q_absorbed[:,h] = q_raw[:,h, 0:128] @ W_uk[h]$ repeated for $h = 0 \dots 127$

this is the absorption: it widens the query 128 -> 512 so the cache can be read in its stored latent form



$grid.x$ -> bq row tiles | $grid.y$ -> 4 tiles of 128 over $kv_rank=512$ | $grid.z$ -> the head index, $0..127$
One head per block means $W_uk[h]$ is staged into shared memory once and reused by every row in the tile.

Step 3a — scores and local softmax numerator

$\text{scores} = (q_{\text{absorbed}} \cdot c_{\text{kv}} + q_{\text{rope}} \cdot \text{pe_cache}) / \text{sqrt}(192)$ then store $\exp(s - m_j)$

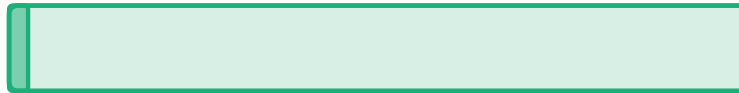
the two dot products (512 and 64 deep) are run as ONE 576-deep reduction

$q_{\text{absorbed}} \mid q_{\text{rope}}$



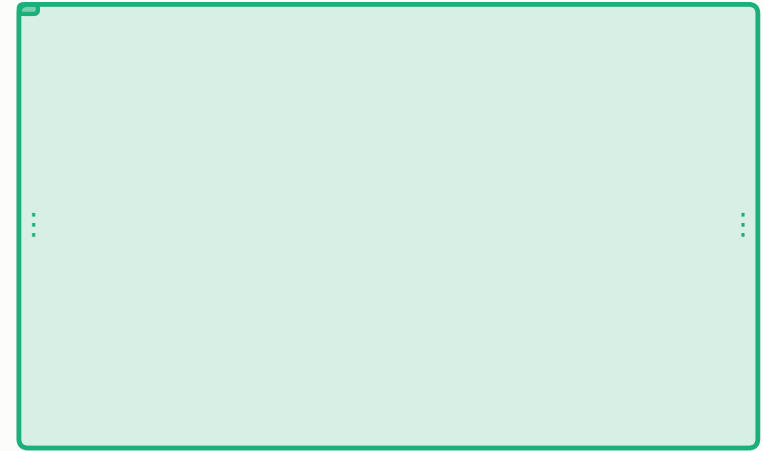
@

$[c_{\text{kv}} \mid \text{pe_cache}]^T$



=

scores



$B \cdot \text{Sq} \cdot H (16,384) \times \text{kv_rank+rope} (576)$

(rows truncated to fit)

128 row tiles -> grid.x

$\text{kv_rank+rope} (576) \times \text{Sk} (4,989)$

39 key panels -> grid.y

$B \cdot \text{Sq} \cdot H (16,384) \times \text{Sk} (4,989)$

(rows truncated to fit)

128 x 128 per block

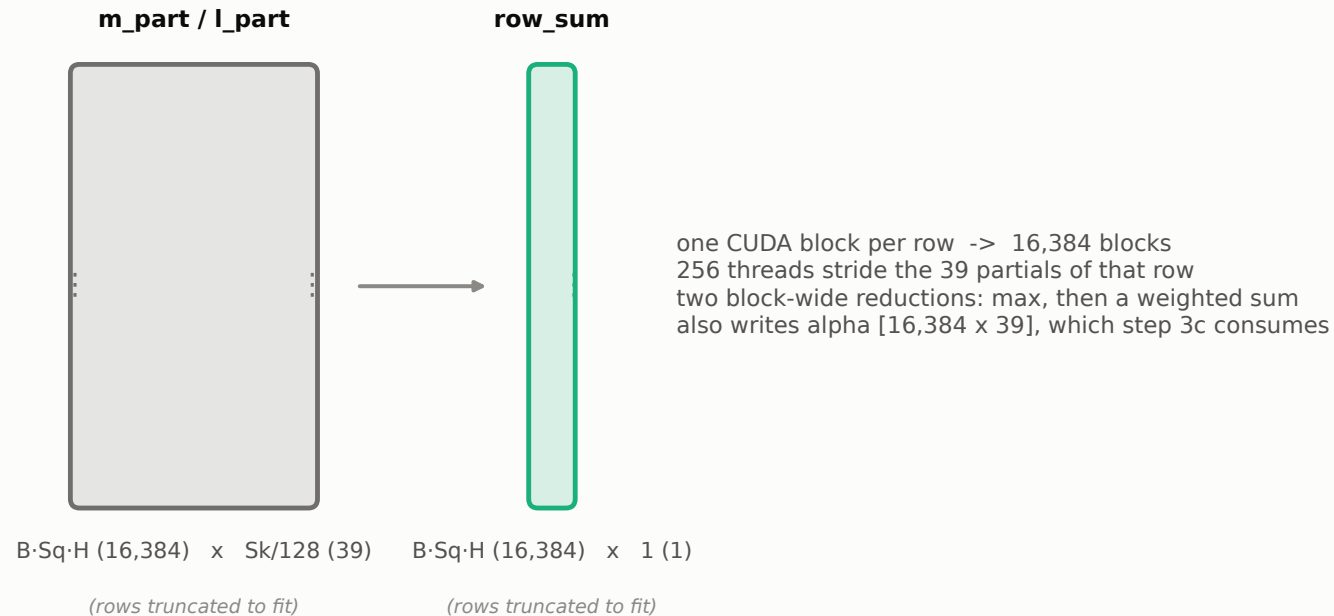
grid.x -> flat = $B \cdot \text{Sq} \cdot H / 128 = 128$ tiles | grid.y -> $\text{Sk} / 128 = 39$ tiles | no grid.z

A block's 128 rows are the 128 heads of ONE query, so the [128 keys x 576] slab it stages -- 512 dims from c_{kv} , then 64 from pe_cache -- is read by all 128 of them.

Step 3b — reconcile the per-block softmax maxima

$$M = \max_j m_j \quad \alpha_j = \exp(m_j - M) \quad \text{row_sum} = \sum_j l_j * \alpha_j$$

3a's grid.y split each row across 39 blocks, none of which could see the global maximum

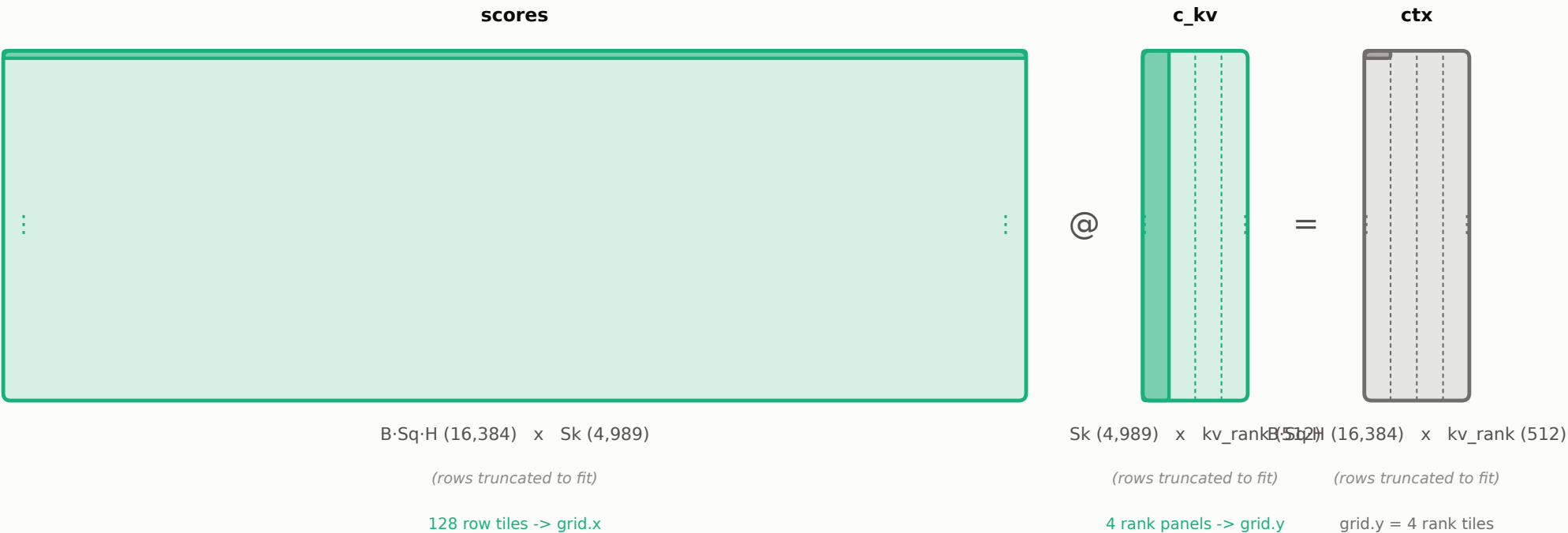


Exact because $\exp(s - m_j) * \exp(m_j - M) = \exp(s - M)$ termwise. Total storage 7.7 MB against a 327 MB scores tensor; costs 0.5% of decode.

Step 3c — context: the attention-weighted sum of the cache

$$\text{ctx} = (\exp(s - m_j) * \alpha_j) @ c_{kv} = \exp(s - M) @ c_{kv}, \text{ unnormalised}$$

alpha_j is folded into the staging load, so what lands in shared memory is already the softmax numerator

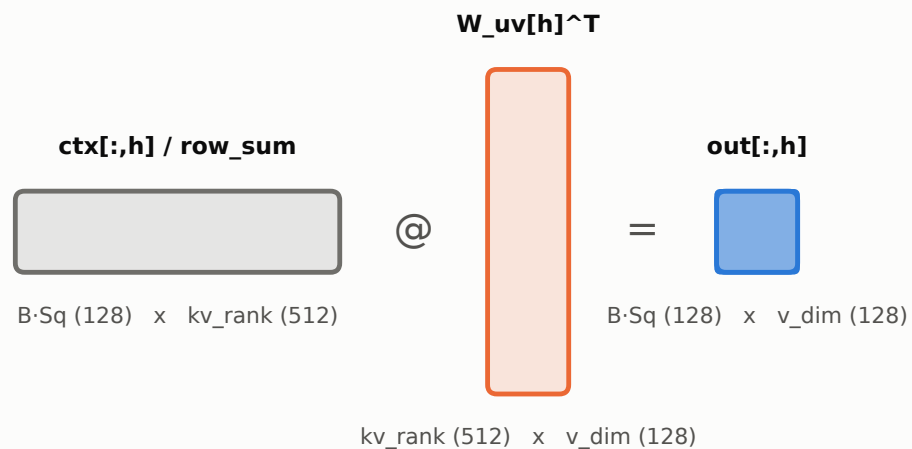


grid.x -> flat / 128 = 128 tiles | grid.y -> 4 tiles over kv_rank = 512 | grid.z -> split-K over Sk
ctx has no Sk axis, so at B=Sq=1 the output alone yields 4 blocks and ~95% of the SMs would idle — hence split-K here.

Step 3d — reconstruct V and project out (once per head)

$\text{out[:,h]} = (\text{ctx[:,h]} / \text{row_sum}) @ W_{uv}[h]^T$ repeated for $h = 0 \dots 127$

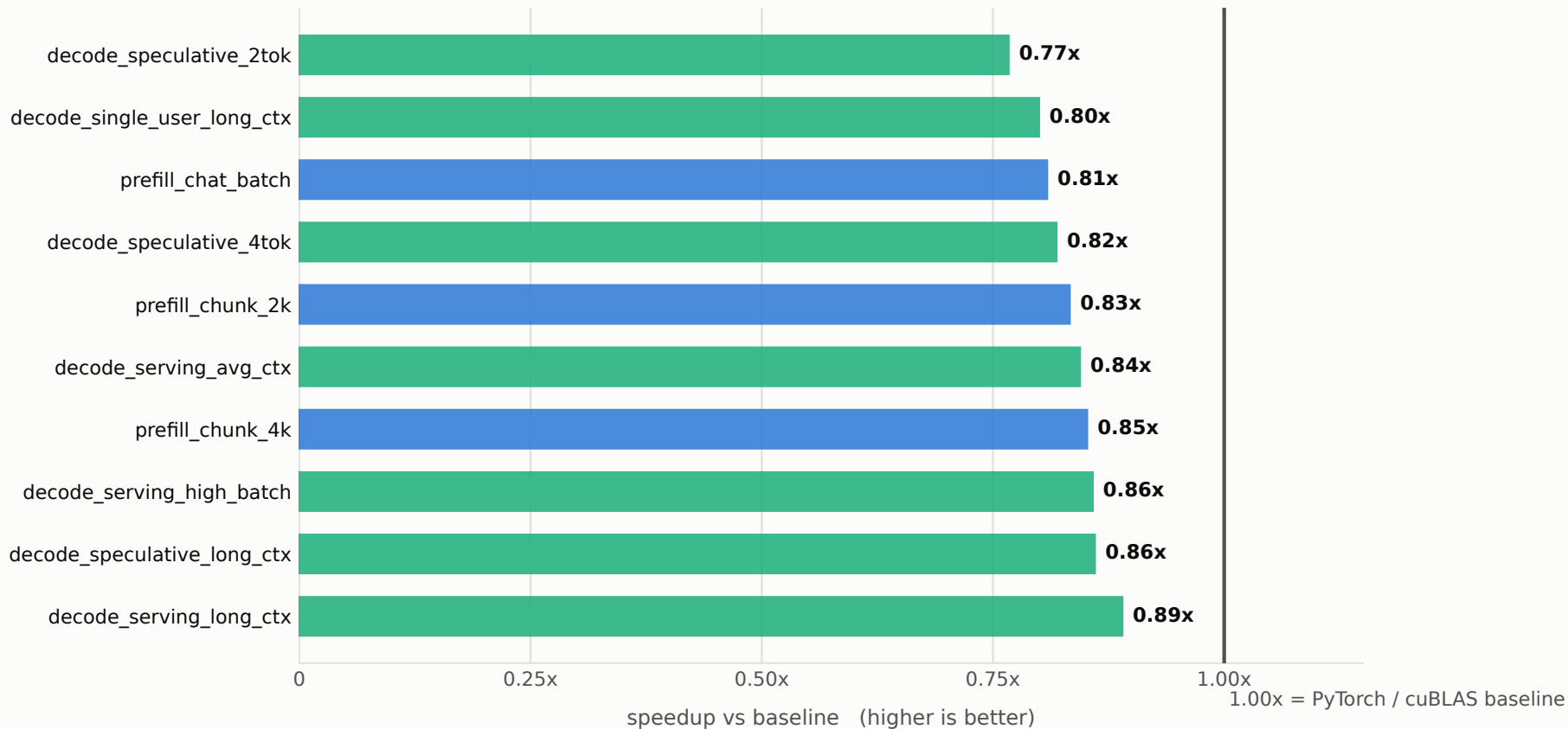
the deferred softmax divide rides along on a staging load this kernel was already doing



grid.x -> bq row tiles | grid.y -> 1 tile over $\text{v_dim} = 128$ | grid.z -> the head index, $0 \dots 127$
Narrows 512 -> 128, undoing the widening that step 2c did. One head per block, same reuse argument as 2c.

End-to-end result: 0.83x geometric mean

FP32, RTX A6000, L2 flushed between iterations, TF32 disabled on both sides. All 10 scenarios pass correctness.



Decode (aqua) and prefill (blue) land in the same narrow band: the tiling generalises, but cuBLAS keeps a ~17% edge.